

PAF ONE-STOP STUDY GUIDE

Comprehensive revision notes + exam-focused concepts + 50 MCQs with answers

Prepared from the uploaded PAF materials and question patterns

Best For

Last-minute revision, MCQ practice, and concept refresh

High-Yield Topics

JS basics, promises, arrays, closures, React hooks, Parcel, APIs

How to Use

Read the cheat sheets first, then attempt the 50 MCQs, then revise weak areas

1. Exam-Focused Topic Map

What the uploaded PAF material strongly suggests

- The paper is heavily MCQ-oriented.
- High-frequency topics include JavaScript fundamentals, arrays, closures, promises, fetch/API usage, React hooks, and frontend tooling.
- The visible questions repeatedly test const reassignment, useReducer, Parcel, useEffect with an empty dependency array, Promise chaining, closures, map(), and JSONPlaceholder.

Concept	What to remember
JavaScript basics	Variables, scope, operators, arrays, objects, functions, callbacks
ES6+	let/const, arrow functions, template literals, destructuring, spread/rest
Async JavaScript	Promises, .then(), .catch(), async/await, fetch
React	Components, props, state, JSX, useState, useEffect, useReducer
Tooling	Parcel bundler, zero-configuration setup, local frontend workflow
API practice	JSONPlaceholder, mock data, response.json(), rendering fetched data

2. JavaScript Core Revision

Variables and scope

- var is function-scoped, while let and const are block-scoped.
- const cannot be reassigned after declaration.

- A const object or array can still have its internal contents changed unless it is frozen.
- Scope means where a variable can be accessed in the program.

Equality and operators

- `==` compares value after type conversion.
- `===` compares both value and type and is usually safer.
- Common logical operators are `&&`, `||`, and `!`.

Functions and callbacks

- Functions can be declared normally or with arrow syntax.
- A callback is a function passed into another function to be executed later.
- Arrow functions are shorter and very common in array methods and React.

High-yield memory line

- If you try to reassign const, JavaScript throws an error.
- If an MCQ asks what a callback is, the safest answer is “a function passed to another function”.

3. Arrays, Objects, and Closures

Essential array methods

Concept	What to remember
<code>map()</code>	Creates a new array by transforming each element.
<code>filter()</code>	Creates a new array with only elements that match a condition.
<code>reduce()</code>	Combines array values into one final value.
<code>forEach()</code>	Runs a function for each element but does not create a new transformed array.
<code>push()</code>	Adds an item to the end of an array.
<code>pop()</code>	Removes the last item from an array.

Closures

- A closure is a nested function that remembers variables from its outer scope.
- This is tested directly in the uploaded PAF material.
- Closures are useful for counters, private state, and callback behavior.

Must-remember example

- `[1, 2, 3].map(x => x * 2)` returns `[2, 4, 6]`.
- A closure remembers outer variables even after the outer function has completed.

4. Asynchronous JavaScript and APIs

Promises

- A Promise represents a value that may be available now, later, or never.
- A Promise can be pending, fulfilled, or rejected.
- `.then()` is used for success handling and `.catch()` is used for error handling.
- A `fetch()` call returns a Promise.

`async` / `await`

- `async` makes a function return a Promise.
- `await` pauses inside an `async` function until the Promise resolves.
- This usually makes asynchronous code easier to read than chained `.then()` calls.

JSON and APIs

- JSON stands for JavaScript Object Notation.
- `response.json()` converts the response body into usable JSON data.
- `JSONPlaceholder` is a fake REST API used for practice and mock data.

Exam shortcuts

- If asked what `fetchData().then(...)` returns, the answer is Promise.
- If asked what `JSONPlaceholder` is used for, the answer is mock or fake API data.

5. React Fundamentals

Core concepts

- React is a JavaScript library for building user interfaces.
- A component is a reusable UI unit.
- Props are used to pass data from parent to child components.
- State stores data that changes over time inside a component.
- JSX lets you write HTML-like syntax inside JavaScript.

Important hooks

Concept	What to remember
<code>useState</code>	Stores local component state such as counters, text input, or toggles.
<code>useEffect</code>	Runs side effects such as API calls, timers, subscriptions, or DOM work.
<code>useReducer</code>	Best for more advanced or complex state logic, especially when updates depend on action types.

MCQ pattern from the uploaded files

- useReducer is used to apply advanced or complex state logic.
- useEffect(() => {}, []) runs only once after the initial render.

6. Tooling and Practical Frontend Knowledge

- A bundler collects and prepares frontend files for the browser.
- Parcel is popular because of its zero-configuration setup.
- In beginner practicals, Parcel is often tested as an easy bundler that works with minimal setup.

One-line memory rule

- Parcel = zero-configuration setup.

7. One-Page Cheat Sheet

Term	Fast memory cue
const	Cannot be reassigned.
Closure	Function remembers outer scope.
map()	Returns a new transformed array.
Promise	Represents a future value.
.then()	Handles a fulfilled Promise and itself returns a Promise.
fetch()	Used to request data from an API and returns a Promise.
response.json()	Parses the response body as JSON.
JSONPlaceholder	Fake API for testing and practice.
useEffect([], ...)	When dependency array is empty, it runs once after first render.
useReducer	Used when state logic is complex.
Parcel	Bundler known for zero-config setup.

8. 50 MCQs with Answers

MCQs 1-10

No.	Question	Answer
1	What happens if you try to reassign a const variable? A. It changes the	B

	value B. It throws an error C. It creates a copy D. It becomes undefined	
2	What is useReducer mainly used for? A. Styling components B. Delaying render C. Complex state logic D. Creating routes	C
3	Parcel is known for: A. Complex setup B. Zero-configuration setup C. Replacing React D. Server hosting	B
4	What does useEffect(() => {}, []) mean? A. Runs on every render B. Runs only once C. Never runs D. Runs before render	B
5	What does .then(...) return? A. String B. JSON C. Object D. Promise	D
6	What is a closure? A. A class method B. A function with no return C. A nested function that remembers outer scope D. A sealed object	C
7	What is the output of [1,2,3].map(x => x * 2)? A. [1,2,3,6] B. [2,4,6] C. 6 D. null	B
8	JSONPlaceholder is mainly used for: A. CSS styling B. DOM editing C. Mock API data D. Data encryption	C
9	Which keyword is block scoped? A. var B. let C. define D. goto	B
10	Which keyword cannot be reassigned? A. var B. let C. const D. function	C

MCQs 11-20

No.	Question	Answer
11	Which operator checks value and type? A. == B. = C. === D. !=	C
12	Which one is an array method? A. map() B. print() C. scan() D. input()	A

13	Which method creates a new transformed array? A. forEach() B. map() C. push() D. pop()	B
14	Which method returns only elements matching a condition? A. filter() B. map() C. join() D. slice()	A
15	Which method combines array values into one value? A. filter() B. reduce() C. map() D. find()	B
16	Which one is used to define a function quickly? A. Arrow function B. While loop C. Switch D. Class	A
17	Which is the correct arrow function syntax? A. => (a,b) B. (a,b) => a+b C. function => a+b D. a,b -> a+b	B
18	What is a callback? A. A loop B. A function passed to another function C. A string method D. A CSS selector	B
19	Which one is used for asynchronous result handling? A. Promise B. Number C. Boolean D. Object literal	A
20	Which promise method handles errors? A. .map() B. .catch() C. .push() D. .forEach()	B

MCQs 21-30

No.	Question	Answer
21	Which keyword is used before a function to use await inside it? A. sync B. async C. promise D. yield	B
22	What does await do? A. Stops forever B. Waits for a promise to resolve C. Converts data to JSON D. Creates a callback	B
23	Which function is commonly used to request API data in JavaScript? A. alert() B. fetch() C. prompt() D. parseInt()	B
24	What does res.json() do? A. Converts response body into JSON data B. Deletes JSON C. Sends CSS	A

	D. Reloads the page	
25	What does JSON stand for? A. Java Source Object Notation B. JavaScript Object Notation C. Java Syntax Output Network D. Joined Script Open Notation	B
26	Which of the following is a JavaScript object? A. "hello" B. 25 C. {name: "Nimal"} D. true	C
27	Which of the following is a valid array? A. {1,2,3} B. [1,2,3] C. (1,2,3) D. <1,2,3>	B
28	What does forEach() usually return? A. A new array B. A number C. undefined D. A Promise	C
29	Which hook is used to store local component state? A. useEffect B. useState C. useRoute D. useClass	B
30	Which hook is used for side effects? A. useState B. useEffect C. useStyle D. useHtml	B

MCQs 31-40

No.	Question	Answer
31	Which hook is better when state updates are complex? A. useMemo B. useReducer C. useRef D. useContext	B
32	A React component should return: A. Only numbers B. JSX or UI output C. Only arrays D. Only CSS	B
33	Props are used to: A. Store database records B. Pass data between components C. Compile JavaScript D. Run SQL queries	B
34	State in React is used to: A. Store changing data B. Design CSS C. Connect Wi-Fi D. Write HTML comments	A
35	Which symbol is commonly used in JSX to embed JavaScript? A. [] B. ()	C

	C. {} D. <>	
36	Which of these is a frontend bundler? A. Parcel B. MySQL C. Oracle D. Python	A
37	Why is Parcel popular for beginners? A. Needs many config files B. Zero-config setup C. Only works offline D. Replaces JavaScript	B
38	Which method adds an item to the end of an array? A. push() B. pop() C. shift() D. map()	A
39	Which method removes the last item of an array? A. push() B. pop() C. unshift() D. concat()	B
40	Which statement is true about let? A. It is function-only scoped B. It is block scoped C. It cannot change D. It is only for arrays	B

MCQs 41-50

No.	Question	Answer
41	Which statement is true about const? A. It must always hold a number B. It can be reassigned freely C. It cannot be reassigned D. It is only for functions	C
42	Which one is used to print something in the browser console? A. console.log() B. document.writeCSS() C. window.printJSON() D. print.console()	A
43	What is the output type of map()? A. Boolean B. New array C. Promise D. String	B
44	Which one is best for fake REST API practice? A. JSONPlaceholder B. Oracle XE C. Mongo shell D. Bootstrap	A
45	What is the main purpose of useEffect? A. To create CSS classes	B

	B. To handle side effects C. To define SQL tables D. To replace components	
46	Which of these is an asynchronous JavaScript feature? A. Promise B. map() C. filter() D. length	A
47	Which one best describes reduce()? A. Removes duplicates only B. Combines array elements into one result C. Creates HTML D. Styles components	B
48	A closure helps mainly because it: A. Deletes outer variables B. Remembers outer variables C. Stops function execution D. Creates database rows	B
49	What is usually returned from fetch()? A. Integer B. Promise C. Array D. Boolean	B
50	Which is the best description of React? A. Database engine B. CSS library C. UI library for building interfaces D. Server operating system	C

9. Answer Key Only

- 1-B, 2-C, 3-B, 4-B, 5-D, 6-C, 7-B, 8-C, 9-B, 10-C
- 11-C, 12-A, 13-B, 14-A, 15-B, 16-A, 17-B, 18-B, 19-A, 20-B
- 21-B, 22-B, 23-B, 24-A, 25-B, 26-C, 27-B, 28-C, 29-B, 30-B
- 31-B, 32-B, 33-B, 34-A, 35-C, 36-A, 37-B, 38-A, 39-B, 40-B
- 41-C, 42-A, 43-B, 44-A, 45-B, 46-A, 47-B, 48-B, 49-B, 50-C

10. Last-Minute Revision Strategy

- First, memorize the one-page cheat sheet.
- Second, answer all 50 MCQs without looking at the key.
- Third, mark weak areas and revise only those sections again.
- Finally, repeat the answer key aloud once before the exam.

Highest-yield final memory list

- const cannot be reassigned.
- useReducer handles complex state logic.

- Parcel is known for zero-configuration setup.
- `useEffect` with `[]` runs once after the first render.
- `.then()` and `fetch()` both keep you in Promise-based flow.
- `JSONPlaceholder` is for fake API testing.
- Closure = function remembers outer scope.
- `map()` returns a new array.